

Assignment 4

1. What happens if you use list comprehension without assigning it to a variable?

- a) Raises a syntax error
- b) It creates a list but doesn't store it
- c) It modifies the original list
- d) It returns a generator object

Answer: b) It creates a list but doesn't store it

2. In a list comprehension, what does the expression represent?

- a) The iterable being traversed
- b) The condition for filtering elements
- c) The value to be included in the new list
- d) The index of the current element

Answer: c) The value to be included in the new list

3. What is the role of if class in list comprehension?

- a) It sets the value of expression
- b) It creates the filter for element
- c) It is used for iteration
- d) It reverses the order of elements

Answer: b) It creates the filter for element

4. Which statement is true about an OrderedDict?

- a) It does not maintain the order of elements
- b) It raises a KeyError if the key is missing
- c) It remembers the order in which keys are inserted
- d) It allows only numerical keys

Answer: c) It remembers the order in which keys are inserted

5. What happens when you try to access a missing key in

defaultdict?

- a) It returns None
- b) It raises a KeyError
- c) It removes the key from dictionary
- d) It returns the default value set for the dictionary

Answer: d) It returns the default value set for the dictionary

6. Which of the following is not part of the collections module?

- a) Counter
- b) OrderedDict
- c) ArrayList
- d) deque

Answer: c) ArrayList

write a program that iterates over elements as many times as its count

```
print("Program that iterates over elements as many times as its count:")
```

```
l=[1,2,3,4,4,2,3,1,3,2,1,2]
```

```
d={}
```

```
for i in l:
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```
    if i in d:
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```
        d[i]+=1
```

```
    else:
```

```
        d[i]=1
```

```
for key,value in d.items():
```

```
    for i in range(value):
```

```
        print(key)
```

```
print("-----")
```

```
-----")
```

```
# write a program that iterates through words/elements
and displays only the element at a given index and
breaks at the third index
print("iterates over specified index and breaks at third
index: ")
s=input("enter a string:")
l=s.split()
ind=int(input("enter the index:"))
for i in range(len(l)):
    if i == ind:
        print(l[i])
    if i ==3:
        break
print("-----
-----")
```

```
# to find the occurence of 10 most common words
print("Occurrence of 10 most common words:")
from collections import Counter
txt=input("Enter text: ").lower().split()
count=Counter(txt)
for word,c in count.most_common(10):
    print(word,"-",c)
print("-----
-----")
```

```
# create a deque and perform append , remove and reverse
operations
print("Operations on deque:")
from collections import deque
d=deque([12,23,34,45])
```

```

print("Original deque: ",d)
d.append(56)
print("After appending: ",d)
d.remove(34)
print("After removing: ",d)
d.reverse()
print("Reverse: ",d)
print("-----")
print("-----")

```

```

# creating a set of duplicates from list
print("Set of duplicate elements of a list:")
l=eval(input("Enter a list:"))
l1=[x for x in l if l.count(x)>1]
s=set(l1)
print(s)
print("-----")
print("-----")

```

```

# total revenue product-wise
print("Product - wise total revenue: ")
sales=[("Rice",1300),
        ("Wheat",2600),
        ("Ragi",900),
        ("Wheat",1100),
        ("Rice",2800)
        ]
d={}
for i in sales:
    if i[0] in d:
        d[i[0]]+=i[1]
    else:
        d[i[0]]=i[1]

```

```
print(d)
```

```
print("-----")
```

```
-----")
```